

Technical Document #822: Deep Learning - Comprehensive Overview

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Attention mechanisms allow models to focus on different parts of the input when producing output. They have become essential in sequence-to-sequence models.

Dropout is a regularization technique where randomly selected neurons are ignored during training. This prevents co-adaptation and improves generalization.

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Convolutional neural networks (CNNs) are designed to process grid-like data such as images. They use convolutional layers to automatically learn spatial hierarchies of features.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Transfer learning is a technique where a model trained on one task is re-purposed on a second related task.
- Convolutional neural networks (CNNs) are designed to process grid-like data such as images.
- Generative adversarial networks (GANs) consist of two neural networks competing against each other.